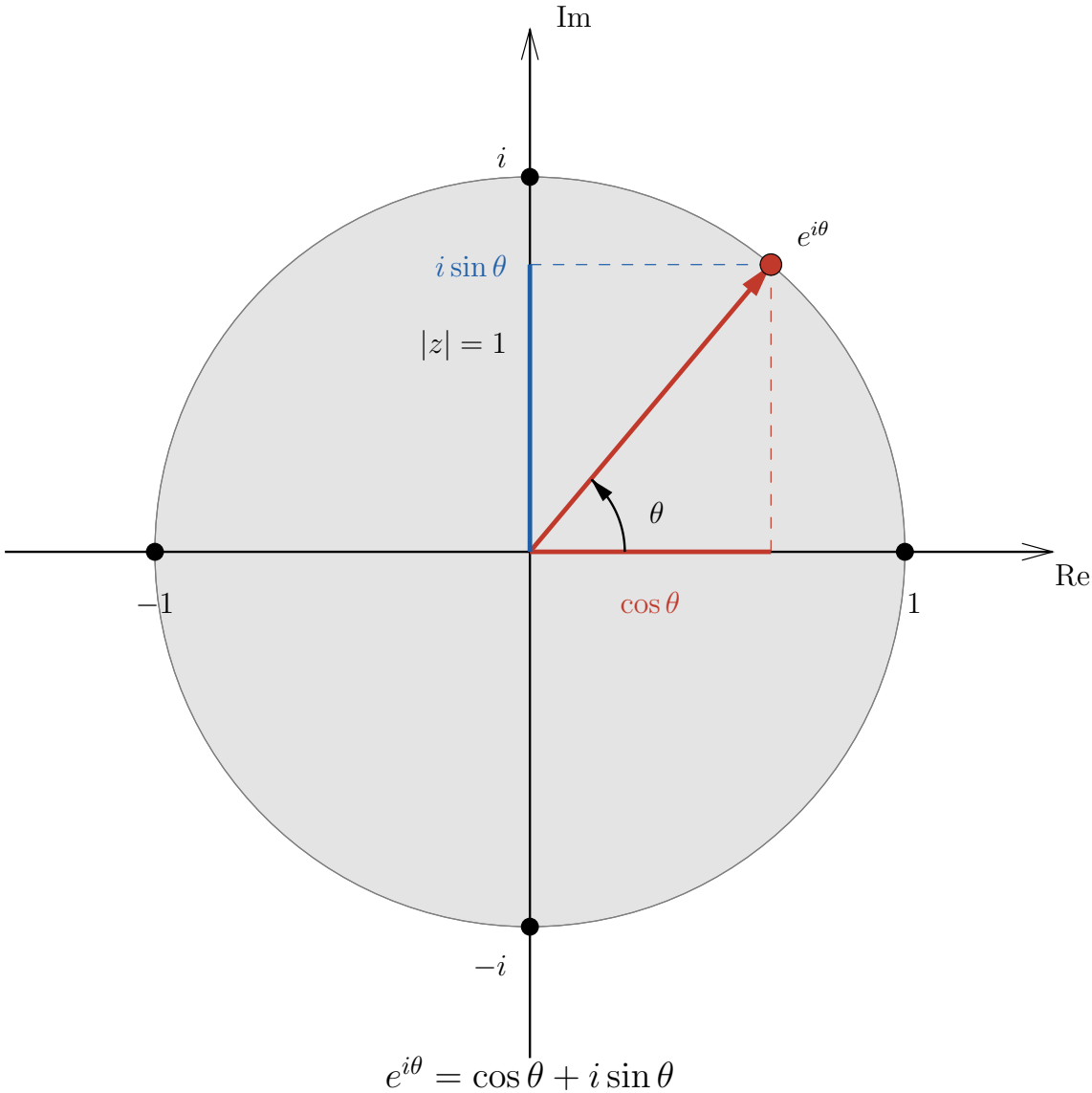
Euler's formula in the complex plane

$e^{i\theta}$ runs around the unit circle; its projections are $\cos\theta$ and $\sin\theta$



Euler's identity:

$$e^{i\pi} + 1 = 0$$